

KASTORIA ARISTOTELIS, GreeceWMO: **166140**Lat: **40.446N**Lon: **21.282E**Elev: **2167**StdP: **13.58**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.0	21.3	7.5	8.8	23.3	11.8	10.8	26.3	21.4	34.6	18.8	35.6	3.5	300	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.4	93.1	64.3	90.1	63.8	87.5	63.2	67.5	84.4	66.0	83.2	64.8	81.8	6.6	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.5	91.7	71.3	60.8	86.3	69.7	59.4	81.9	69.5	33.1	84.8	31.9	83.3	30.9	81.7	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.4	14.6	12.6	DB	10.4	96.8	6.1	2.7	5.9	98.8	2.4	100.4	-1.1	101.9	-5.5	103.8	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.6	35.1	39.9	45.6	52.8	60.3	68.4	73.0	73.6	65.5	56.0	46.0	38.3	(6)	
	DBStd	14.37	7.30	6.34	6.64	6.92	4.87	4.95	4.39	3.64	5.66	5.60	6.51	6.15	(7)	
	HDD50	1492	462	285	167	49	2	0	0	0	0	15	147	365	(8)	
	HDD65	4522	926	703	602	368	157	23	3	0	63	281	569	827	(9)	
	CDD50	3181	0	2	30	132	321	553	714	732	465	201	28	3	(10)	
	CDD65	738	0	0	0	1	11	126	252	267	78	3	0	0	(11)	
	CDH74	8584	0	0	5	75	223	1507	2882	2962	878	52	0	0	(12)	
	CDH80	3359	0	0	0	6	26	504	1255	1302	260	6	0	0	(13)	
(14) Wind	WSAvg	4.1	3.5	3.8	4.8	4.6	4.4	4.8	5.2	4.3	4.0	3.2	2.8	3.5	(14)	
(15) Precipitation	PrecAvg	22.5	1.8	1.8	1.8	1.9	1.9	1.3	1.0	1.1	1.9	2.6	2.8	2.6	(15)	
	PrecMax	34.6	4.4	4.7	4.1	4.7	4.1	3.0	2.6	3.5	6.4	5.9	6.2	6.3	(16)	
	PrecMin	14.0	0.0	0.2	0.3	0.6	0.3	0.3	0.0	0.0	0.3	0.0	0.8	0.0	(17)	
	PrecStd	5.5	1.2	1.3	0.9	1.0	0.9	0.8	0.7	0.7	1.7	1.6	1.3	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.8	64.9	72.3	80.7	82.7	91.8	97.9	96.7	89.8	79.3	68.0	58.7	(19)	
		MCWB	46.1	49.0	51.2	56.8	59.6	65.1	67.1	63.5	62.8	60.8	53.5	49.4	(20)	
	2%	DB	52.2	58.8	66.8	75.9	79.2	88.6	93.7	93.1	86.2	74.8	63.5	54.8	(21)	
		MCWB	44.2	46.9	49.1	54.6	59.5	64.2	64.9	63.7	61.4	57.5	51.9	45.8	(22)	
	5%	DB	48.8	54.6	62.9	70.7	76.3	85.8	90.4	90.0	83.0	71.4	59.4	51.9	(23)	
		MCWB	42.5	44.5	47.9	52.9	58.9	63.3	64.4	63.2	61.0	56.5	50.5	45.2	(24)	
	10%	DB	45.5	51.1	58.6	66.4	73.3	82.8	87.3	87.3	79.6	67.5	55.9	49.1	(25)	
		MCWB	41.3	43.2	46.2	51.9	57.6	62.3	64.0	62.6	60.2	55.3	49.4	42.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.7	50.6	53.8	57.8	63.5	68.0	71.4	68.0	69.9	64.3	56.4	51.8	(27)	
		MCDB	54.2	61.0	68.6	76.2	76.0	84.0	92.2	85.6	80.5	74.6	61.4	54.5	(28)	
	2%	WB	45.6	47.9	50.8	56.0	61.6	66.3	67.9	66.4	64.4	60.6	53.8	48.9	(29)	
		MCDB	50.1	57.0	63.4	71.6	74.7	83.2	86.3	84.0	77.1	69.0	60.3	52.6	(30)	
	5%	WB	43.6	45.9	48.7	54.3	60.1	64.8	66.2	65.2	62.7	58.5	52.2	45.7	(31)	
		MCDB	47.7	53.1	60.0	67.3	73.2	81.2	84.1	82.8	75.9	65.9	57.6	50.8	(32)	
	10%	WB	41.8	44.0	46.9	52.6	58.7	63.5	64.9	64.1	61.5	57.0	50.5	43.2	(33)	
		MCDB	45.2	50.2	57.5	64.0	70.7	79.5	82.7	81.8	73.9	63.7	55.2	47.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.6	16.1	18.6	21.0	21.8	24.2	25.4	26.6	22.2	18.5	16.4	16.5	(35)	
		MCDBR	20.1	24.8	28.1	29.2	28.0	29.4	31.2	31.2	29.5	26.6	23.0	21.4	(36)	
		MCWBR	13.0	14.2	13.3	11.9	11.1	9.6	9.4	8.4	9.6	11.7	13.1	14.0	(37)	
	5% WB	MCDBR	17.2	22.3	25.2	26.1	25.5	27.3	28.0	26.5	23.8	20.3	18.8	18.0	(38)	
		MCWBR	11.7	13.2	12.7	10.9	11.2	10.2	9.8	8.2	9.0	10.0	11.4	11.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.326	0.356	0.404	0.465	0.461	0.439	0.447	0.459	0.440	0.396	0.355	0.327	(40)	
	taud		2.385	2.277	2.161	2.000	2.069	2.183	2.161	2.124	2.144	2.259	2.339	2.374	(41)	
	Ebn at Noon		262	269	266	257	261	267	263	256	252	250	247	251	(42)	
	Edh at Noon		27	34	43	54	52	46	47	47	43	35	28	25	(43)	
(44) All-Sky Solar Radiation	RadAvg		582	825	1174	1506	1820	2104	2202	1950	1438	1012	671	513	(44)	
	RadStd		78	110	136	166	123	154	122	121	118	107	67	90	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (15 neighbors)		+0.72	N/A	N/A	+1.28	N/A	N/A	N/A	-180	+180	N/A	

Nomenclature: See separate page